

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Soft Computing

Course Code: CSPC-612(vi)

Examination: 2nd Periodical Examinations

Scheme: NEP

Branch: CSE (AI & ML)

Semester: VI

Question Paper for 2nd Periodical Examination (May 2026) (Session Jan-May 2026)

Time: One and a half hours

Max Marks:30

Note: Attempt all questions

CO	COURSE OUTCOMES
CO1	Explain the fundamental principles of soft computing and analyze the structure and working of artificial neural networks.
CO2	Apply various neural network models, architectures and learning methods to solve basic computational problems.
CO3	Develop and analyze fuzzy logic-based systems using fuzzy sets, inference mechanisms and defuzzification techniques.
CO4	Apply supervised, unsupervised and reinforcement learning techniques to solve real-world data analysis problems.

Que No.	Questions	Marks	Target CO	Target KL
1.a	What type of models are GPT models?	1	CO4	L1
1.b	What is the key mechanism used in Transformer?	1		
1.c	What architecture is used on GPT?	1		
1.d	Which metric measures signal quality in generated data by GAN.	1		
1.e	What type of output do generative models produce?	1		
2.	Describe the architecture of a Neuro- fuzzy system. Explain the function of its layers.	5	CO3	L2
3.	Explain the architecture and working of the Generative Adversarial Networks.	5	CO4	L2
4.	Explain the architecture and working of the Transformer.	5	CO4	L2
5.	Explain the concept of Defuzzification in a fuzzy logic system. Apply any three defuzzification methods to a suitable example and show how a crisp output is obtained.	10	CO3	L3

KL: Knowledge Level: L1 – Remember L2 – Understand L3 – Apply L4 Analyze L5 – Evaluate L6 – Create
Course Advisor: Er. Aditi Mahajan (Assistant Professor)